# Algorithms as Recipes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

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TEACHER GUIDE · 1 OF 9

## Algorithms as Recipes

> Prepare Algorithms as Recipes: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: **Trace** an ordered **algorithm** using the current value after every instruction

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Features: Useful Clues

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

Look for: Choose recorded properties that preserve distinctions needed for a **state**d task

Reference example, not the only acceptable wording: Use length and width measured before the decision. **State** the order as length in cm, then width in cm; the record is 12, 8.

The target is fits or too large under the guide, and this card is fits because both equalities are included.

Colour and card number are not needed for this rule; the final inspection sticker copies the answer and is unavailable before inspection.

### Sound Becomes a Signal

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

Look for: Interpret an ordered **sequence** of sound-signal samples with a **state**d time interval

Reference example, not the only acceptable wording: Five samples, a 2 ms interval, and 8 ms from first to last. Consecutive pairs make gaps 0-2, 2-4, 4-6 and 6-8: four gaps between five samples.

### Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

Look for: Describe a measurement using its quantity, value, unit and conditions

Reference example, not the only acceptable wording: Keep the observed value 24 and approximate time lunchtime; mark the unit, exact time if needed, location, exposure and device identity as unknown or to be checked.

A suitable completed record needs air temperature, value, unit and relevant conditions. Invented degrees C or shade is not an **observation**.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson **sequence**.

### What the learner must demonstrate

> On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. **Trace** position and direction after every instruction. What if TURN were unspecified?

### Concept to explain

An **algorithm** is an ordered, testable procedure, with or without code.

| Term | Meaning |
| --- | --- |
| algorithm | ordered procedure for a task |
| step | one executable action |
| sequence | the order in which steps run |

| Term | Meaning |
| --- | --- |
| assumption | condition taken for granted |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

### Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

This week develops precise sequential execution with small whole-number values. An **algorithm** is a procedure; **state** consists of the current values being tracked. Make the transition from one **variable** to two explicit.

Copying in these exercises means copying a number value, not linking boxes. Avoid presenting this as a claim about every programming-language object model.

No software accounts are required, and a fixed arithmetic procedure is not **evidence** of learning from data.

### Readiness answer

The selected record hides width, which this task needs. Identical length-only **input**s can require different labels. We must retain the relevant information instead of expecting a later **process** to recover it with certainty.

An erasable box labelled Score and instruction cards for add, double, subtract, set, report and stop

Two erasable boxes labelled X and Y, and paper **trace**s with Stage, X and Y columns

The independent W4 task on a separate sheet; counters may support addition before students **trace** without them

**End goal for the pictorial case**

By the end, I can **trace** changing **state** and explain why instruction order matters.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain the mechanism

> Model Algorithms as Recipes using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

**1. Input — information entering**

**INPUT** goal, starting **state**, and available objects

**2. Process — the deciding step**

**PROCESS** follow ordered unambiguous **step**s

**3. Output — the result**

**OUTPUT** completed task or a visible failure point

### Key idea

CORE IDEA An **algorithm** is an ordered, testable procedure, with or without code.

### How the mechanism works

UNDER THE HOOD An **algorithm** is a finite procedure whose **step**s and control flow can be executed. Correctness depends on **assumption**s, preconditions, order, and termination.

Code is one representation of an **algorithm**, not the definition itself.

Draw a fresh example with all three stages:

Every **step** connects a claim to observable **evidence**.

### The case to investigate

CASE A paper robot cannot infer missing intentions; it needs ordered **step**s that another person can execute and test.

### Step 1

**STEP** 1 — OBSERVE goal, starting **state**, and available objects

### Step 2

**STEP** 2 — **TRACE** follow ordered unambiguous **step**s

### Step 3

**STEP** 3 — CONCLUDE completed task or a visible failure point

### Model answer

MODEL REASONING A strong **algorithm state**s the starting arrangement, names each object, orders actions, and includes checks such as whether the bag is open.

The revision targets the failed **assumption** rather than replacing every **step**.

### A claim to avoid

DO NOT OVERCLAIM A goal is not yet an **algorithm**; each **step** must tell the actor what to do.

What new **evidence** could change this answer?

**Concrete teaching case**

### The teaching case

A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

### The teaching case

**Trace** the position after each command.

**Exact explanation to model**

### Why the result follows

Middle, rightmost, unchanged. Direction, distance and starting conditions must be specified.

**Explain the added worked example**

## Explain the added worked example · 1

A classroom score display starts with the number 4. Its instructions are: add 3 to score; double score; subtract 2 from score; report score and stop. What number should it report?

Before doing the arithmetic, name the value being tracked and write its starting value. Here the **variable** is score and the **initial state** is score 4.

## Explain the added worked example · 2

An **algorithm** gives precise instructions for a task. In this chapter each instruction is carried out once, in the written order. An update replaces the current value. After adding 3, score is 7.

The next instruction therefore doubles 7, not the original 4. Keeping only the starting value in mind is a common source of errors.

## Explain the added worked example · 3

A **variable** name stays the same while its value may change. Think of score as the label on an erasable number box. Updating the box does not keep all earlier numbers inside it.

A separate **trace** preserves those earlier **state**s so we can explain how the result was reached. This score procedure follows fixed arithmetic instructions; it does not learn from examples.

Write the **initial state** first. For each instruction, read the latest value, perform the operation, and record the new **state**. A transition is the change from one recorded **state** to the next.

## Explain the added worked example · 4

In words, double score means calculate twice its current value and store that result in score. Add 3 to score means increase the current value by 3. Set score to 3 means replace it with 3 instead.

These instructions sound similar but do different things: if score is 7, adding 3 produces 10, while setting it to 3 produces 3.

A **trace** is **evidence** of one execution from one **initial state**. To compare two procedures fairly, keep the **initial state** the same and list every intermediate value.

Then you can point to the first difference instead of only comparing the final numbers.

### Key terms

| Term | Meaning |
|---|---|
| Algorithm | A precise procedure for carrying out a task |
| Variable | A named place for a value that can be updated |
| State | The current values being tracked at a particular point in a process |
| Initial state | The values before the first instruction is carried out |
| Trace | A record of the state as instructions are carried out in order |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding **step**.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Stage from score 4 | Add then double | Double then add |
|---|---|---|
| After first instruction | Add 3 gives 7 | Double gives 8 |
| After second instruction | Double gives 14 | Add 3 gives 11 |
| After subtracting 2 | Report 12 | Report 9 |

Both procedures use add 3, double and subtract 2, but the first two operations are exchanged. Doubling after the addition also doubles the extra 3. Adding afterward does not.

The final difference comes from the order, even though both procedures started at 4.

## Model the reasoning aloud · 5

Now use two number boxes: A starts at 2 and B starts at 0. Copy the current value of A into B, then add 5 to A. After the copy, A is 2 and B is 2. After the addition, A is 7 and B remains 2.

Copying stores the value at that moment; it does not create a lasting connection between these number boxes. Record both values after each instruction, including the one that did not change.

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual **step**; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Algorithms as Recipes

A fictional paper badge machine transforms an initial score using an ordered recipe.

### The facts we will use

#### Supplied facts

Start score=3. Recipe A: double, add 4, subtract 1, report.

#### Supplied facts

Recipe B swaps the first two operations but keeps the same starting score.

AI ACADEMY / AI-1/9

# Changing instruction order

| | Start | After step 1 | After step 2 | Report |
|---|---|---|---|---|
| Recipe A | 3 | 6 | 10 | 9 |
| Recipe B | 3 | 7 | 14 | 13 |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/9

# Algorithms as Recipes

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Start

Both runs begin at 3.



### 2. Recipe A

3 -> 6 after doubling -> 10 after adding 4 -> 9 after subtracting 1.



### 3. Recipe B

3 -> 7 after adding 4 -> 14 after doubling -> 13 after subtracting 1.



### 4. Compare outputs

Same operations, different order: outputs 9 and 13.

**CHANGE ONE THING**

Change the starting score to 0 for both recipes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Start | Both runs begin at 3. |
| 2. Recipe A | 3 → 6 after doubling → 10 after adding 4 → 9 after subtracting 1. |
| 3. Recipe B | 3 → 7 after adding 4 → 14 after doubling → 13 after subtracting 1. |
| 4. Compare outputs | Same operations, different order: outputs 9 and 13. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

Every instruction reads the current score, not the original one unless explicitly told to. In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling.

The final difference of 4 follows from that order, not from different starting data.

### What this example does not establish

#### Keep the stated limit

One **trace**d **input** checks one run; it does not establish every recipe is equivalent or validate an entire program.

### Change one thing in the pictured case

#### Change one thing

Change: Change the starting score to 0 for both recipes.

#### Predicted result

**Prediction**: A reports 3 and B reports 7.

#### Why the result follows

Reason: A: 0→0→4→3. B: 0→4→8→7.

### Ask while showing the picture

## Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next **step**. Uncover one **step**, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Algorithms as Recipes, reduce support deliberately and check individual reasoning before discussing answers.

**Board sequence**

### 1. Put the facts on the board

Write the given facts: A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

### 2. State the decision to explain

Write the question to explain: **Trace** the position after each command.

### 3. Reveal the worked reasoning

Reveal after **prediction**s: Middle, rightmost, unchanged. Direction, distance and starting conditions must be specified.

### Hold back the independent task

Keep for an independent attempt: Instructions say pick up a pencil, write your name, then find paper. Find the first ordering problem and rewrite the three **step**s so another learner can follow them.

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

#### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one **step** at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, **trace** conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

The guided **trace** is 5,7,14,13. Double reads 7 because the add-2 instruction has already replaced 5 with 7.

For the exit response, compare the **initial state**s and the meaning/order of the instructions, then compare successive recorded **state**s. Different starts can give different correct results.

With matching starts and instructions, the first differing transition identifies where to inspect the calculation or interpretation.

### Case-specific teaching sequence

| Stage | Teacher action |
| --- | --- |
| Session 1 start 0 to 7 minutes | Recall why a record must retain relevant information. Introduce score as a named box and write the initial value before the first update. |

| Stage | Teacher action |
| --- | --- |
| 7 to 20 minutes | Model score 4 through add 3, double and subtract 2. Erase and replace the current value while retaining every old value in a separate trace. |
| 20 to 33 minutes | Compare the reordered procedure from the same initial 4. Contrast set score to 3 with add 3 to score using a current value of 7. |
| 33 to 45 minutes | Students complete W1 and W2. Ask them to identify the value each instruction reads, not just announce the final result. |
| Session 2 start 0 to 8 minutes | Model the A and B copy example: 2,0 becomes 2,2 then 7,2. Record unchanged values as well as updated ones. |
| 8 to 23 minutes | Use W3 and the guided response to identify the first incorrect transition. Require later states to be recomputed from the repaired value. |
| 23 to 38 minutes | Collect W4 independently. Learners trace both instruction orders from the same start and explain why copying a number is not a lasting link. |
| 38 to 45 minutes | Discuss W5 and the exit response. Show why an intermediate trace can distinguish procedures whose final values happen to match. Preview conditional instructions. |

### Reduce help deliberately

#### Completion practice

Completion task: Two procedures start score at 6. Procedure A sets score to 4, then adds 2. Procedure B adds 4 to score, then adds 2. **Trace** both and explain why the first instructions are different.

#### Reasoning to listen for

Teacher check: A **trace**s 6, 4, 6. B **trace**s 6, 10, 12. Setting replaces the current value with 4; adding increases the current 6 by 4. The procedures therefore have different intermediate and final **state**s.

Start by identifying the given **input**s together. Leave the intermediate **step**s blank. If needed, model one **step**, then withdraw that help on the next **step**. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Algorithm**s and changing **state**. Proposed explanation: “Every instruction uses the original number”. Decide whether this explanation is supported.

Point to the **step** or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: Point to the most recent **trace** entry. Double after add reads the updated number, not the initial one.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant **step** for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Use the arrows to explain where the two zero-start runs first differ.

### Reasoned teacher answer

They first differ after operation one: A has 0 and B has 4. Their reports are 3 and 7.

Changed-case reasoning: A reports 3 and B reports 7. A: 0→0→4→3. B: 0→4→8→7.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding **step** and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

# Run the weekly evidence lab

> Run the Algorithms as Recipes investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

### 1. Predict

**State** your **prediction** before the result is known.

### 2. Investigate

Complete the task: Write directions for a paper robot to pack a bag.

### 3. Compare cases

Keep a normal case and an edge case clearly separated.

### 4. Record honestly

Record what happened—even when it does not match your **prediction**.

### 5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

### Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

### Find the unsupported claim

CLAIM TO REPAIR Correct actions always work in any order.

For guessing, return to given facts and request a **trace**. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

Every instruction uses the original number: Point to the most recent **trace** entry. Double after add reads the updated number, not the initial one.

Set and add mean the same thing: Start at 6: setting to 4 gives 4 while adding 4 gives 10. Ask which value remains in the box.

Copying keeps number boxes linked: Copy A 2 into B, then change only A to 7. B still contains the copied number 2.

A matching final value proves the whole **process**: Compare 10,12,10 with 10,14,10. The final values match but the **step**s do not.

### Match the support to the first error

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or **assumption** and asking them to defend the effect, not simply by assigning more of the same questions.

## Assess and explain the answers

> Use the Algorithms as Recipes questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

> **Teacher key · Worked case answer**
>
> Middle, rightmost, unchanged. Direction, distance and starting conditions must be specified.

### Independent question

> **Independent question · withhold the key**
>
> Instructions say pick up a pencil, write your name, then find paper. Find the first ordering problem and rewrite the three **step**s so another learner can follow them.

### Expected independent answer

> **Teacher key · Expected independent answer**
>
> Find paper before writing. One valid order is find paper, pick up a pencil, write your name on the paper. Instructions must supply needed materials before the action that uses them.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “Correct actions always work in any order.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

Positions are 1 initially, then 3, 3, 2, 2. Direction changes from increasing to decreasing at TURN.

Without TURN's meaning the later position is ambiguous; an **algorithm** requires enough detail for another person to reproduce it.

### Explanations for the additional workbook examples

### W1 Trace the current value

#### Workbook question · W1

Start score at 3. Add 4 to score, double score, then subtract 5. Record the **initial state** and the **state** after each instruction. Underline the number used by the doubling **step** and explain where it came from.

### Teacher reasoning · W1

The **trace** is 3, 7, 14, 9. The doubling **step** uses 7, the current value after adding 4 to 3. It must not double the initial 3. Accept a table, arrows or clearly labelled sentences that preserve the order.

## W2 Distinguish replacement from addition

### Workbook question · W2

Two procedures start score at 6. Procedure A sets score to 4, then adds 2. Procedure B adds 4 to score, then adds 2. **Trace** both and explain why the first instructions are different.

### Teacher reasoning · W2

A **trace**s 6, 4, 6. B **trace**s 6, 10, 12. Setting replaces the current value with 4; adding increases the current 6 by 4. The procedures therefore have different intermediate and final **state**s.

## W3 Repair the first error

### Workbook question · W3

A procedure starts score at 6, adds 2, doubles score, then subtracts 3. A **trace** says 6, 8, 12, 9. Identify the first wrong transition, explain the likely mistake and write a complete corrected **trace**.

Why is changing only the last value inadequate?

### Teacher reasoning · W3

The transition from 8 to 12 is the first error. It appears to double the initial 6 instead of the current 8. The correct **trace** is 6, 8, 16, 13.

Changing only the last value leaves the incorrect update unexplained and does not show a valid execution.

### W4 Trace two changing number boxes

#### Workbook question · W4

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes.

Then swap the first two instructions and **trace** the new procedure from the same **initial state**.

#### Teacher reasoning · W4

Original **trace** in X,Y order: 4,1 then 4,4 then 7,4 then 7,8. Reporting leaves 7,8 unchanged. Copying captured X as 4; it did not link Y to later changes in X.

With the first two instructions swapped: 4,1 then 7,1 then 7,7 then 7,14. The changed order makes the copy capture 7 instead of 4.

### W5 Test what a final answer can prove

#### Workbook question · W5

The required procedure starts score at 10, adds 2 and subtracts 2. A learner instead adds 4 and subtracts 4. Both report 10.

Write both **trace**s and explain whether matching the final result proves that the learner followed the required procedure.

### Teacher reasoning · W5

Required **trace**: 10,12,10. Learner **trace**: 10,14,10. Matching the final value does not prove the required procedure was followed; the intermediate **state**s and instructions differ.

Inspecting the **trace** reveals the discrepancy even though the final result matches.

## Assess independent transfer

### Independent question · withhold the key

Selected case: For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction.

Explain why Y does not become 7 when X changes. Then swap the first two instructions and **trace** the new procedure from the same **initial state**.

## Selected case answer

### Teacher key · Selected case answer

Original **trace** in X,Y order: 4,1 then 4,4 then 7,4 then 7,8. Reporting leaves 7,8 unchanged. Copying captured X as 4; it did not link Y to later changes in X.

With the first two instructions swapped: 4,1 then 7,1 then 7,7 then 7,14. The changed order makes the copy capture 7 instead of 4.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

#### Feedback that the learner uses

Name the earliest unsupported **step** and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Algorithms as Recipes to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

#### Helpful Classroom Sorter

Write the first executable sorting **algorithm** for one easy item before adding branches or AI.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

#### Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 10: If/Then Decisions. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Use a finger or token to mark the current instruction and provide a blank **trace** with one row per update. Let learners verbalise Read, calculate, replace, record. Allow oral explanations alongside the table.

Extension: starting at 0, compare add 2 then double with double then add 2; explain why their difference remains 2 for any starting whole number.

W4 is secure when both **trace**s include both **variable**s after each instruction, copying preserves the then-current value, and the swapped procedure correctly ends at X 7 and Y 14.

A correct final pair without intermediate **state**s is incomplete **evidence**. After support, retry from X 3 and Y 0 with copy X into Y, add 2 to X, double Y. The expected **state**s are 3,0 then 3,3 then 5,3 then 5,6.

Week 10 adds conditions and boundary cases. Students will decide which instruction runs by testing the current **state**, then **trace** the selected update.

### Plan the delayed return

Suggested return points in this level: ai-1/10, ai-1/12, ai-1/15. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next **step**. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Algorithms as Recipes lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: **Trace** an ordered **algorithm** using the current value after every instruction

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Two procedures start score at 6. Procedure A sets score to 4, then adds 2. Procedure B adds 4 to score, then adds 2. **Trace** both and explain why the first instructions are different.

### Supported practice answer

Reasoned practice answer: A **trace**s 6, 4, 6. B **trace**s 6, 10, 12. Setting replaces the current value with 4; adding increases the current 6 by 4. The procedures therefore have different intermediate and final **state**s.

Check every learner before discussion: Use the worked case for **Algorithm**s and changing **state**. Proposed explanation: “Every instruction uses the original number”. Decide whether this explanation is supported.

Point to the **step** or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Point to the most recent **trace** entry. Double after add reads the updated number, not the initial one.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding **step**. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Change the starting score to 0 for both recipes. Require a **prediction** and a reason before comparing. Expected result: A reports 3 and B reports 7. Reason: A: 0→0→4→3. B: 0→4→8→7.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
| --- | --- | --- | --- |
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Make a transparent first version. Keep a **trace** of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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TEACHER GUIDE · 8 OF 9

## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

> Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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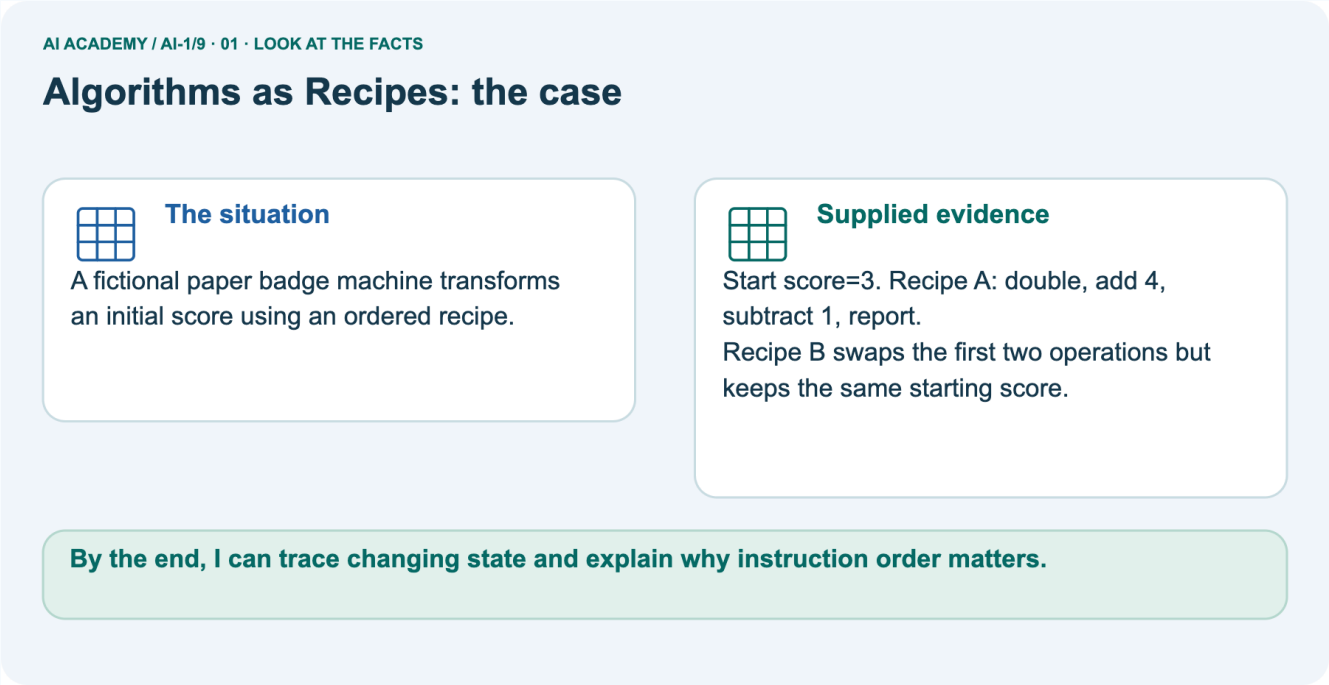


## Visual concept journey

> Show one Algorithms as Recipes graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Algorithms as Recipes: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper badge machine transforms an initial score using an ordered recipe. Supplied facts: Start score=3. Recipe A: double, add 4, subtract 1, report.

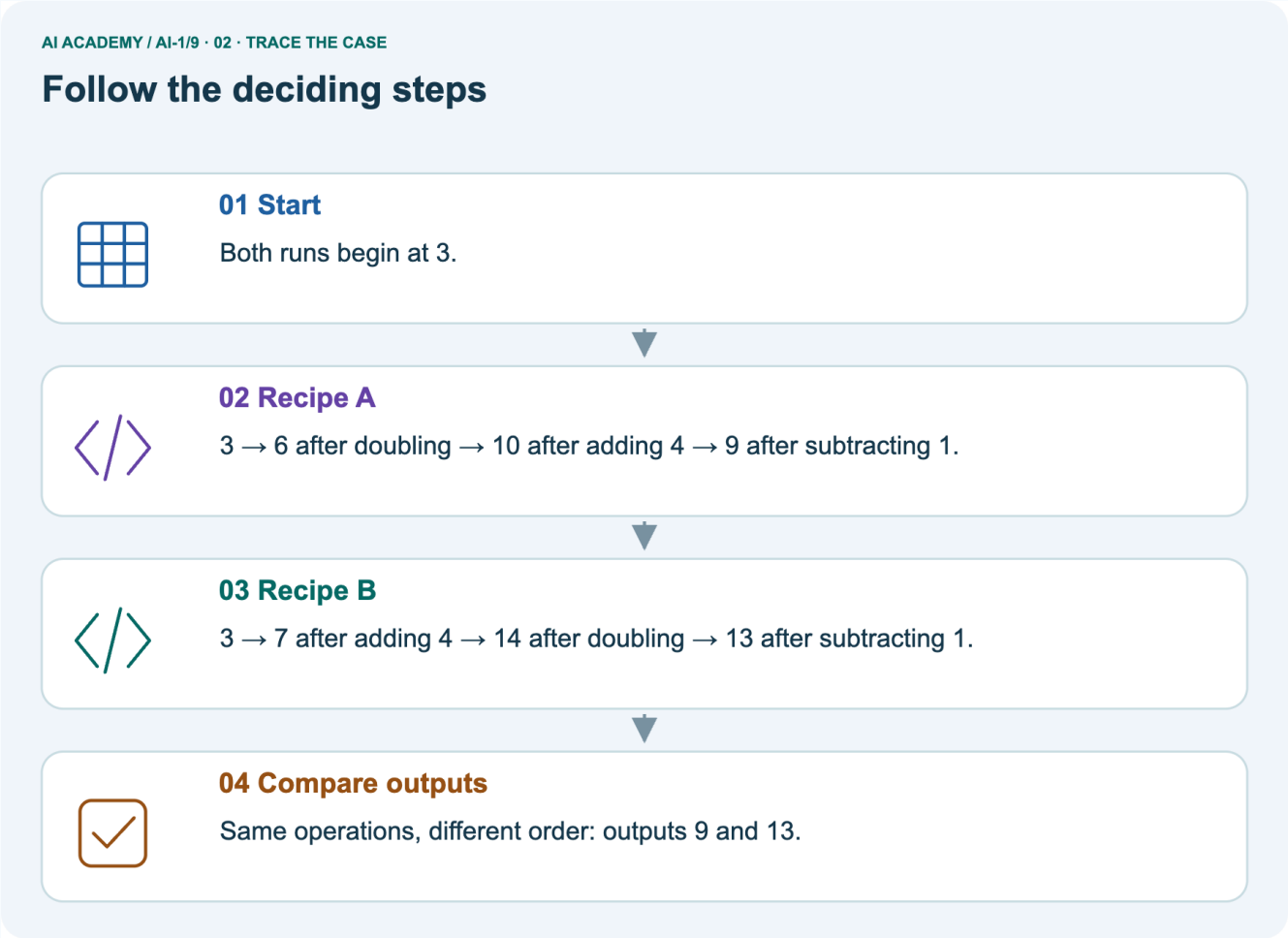
Recipe B swaps the first two operations but keeps the same starting score.

> Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

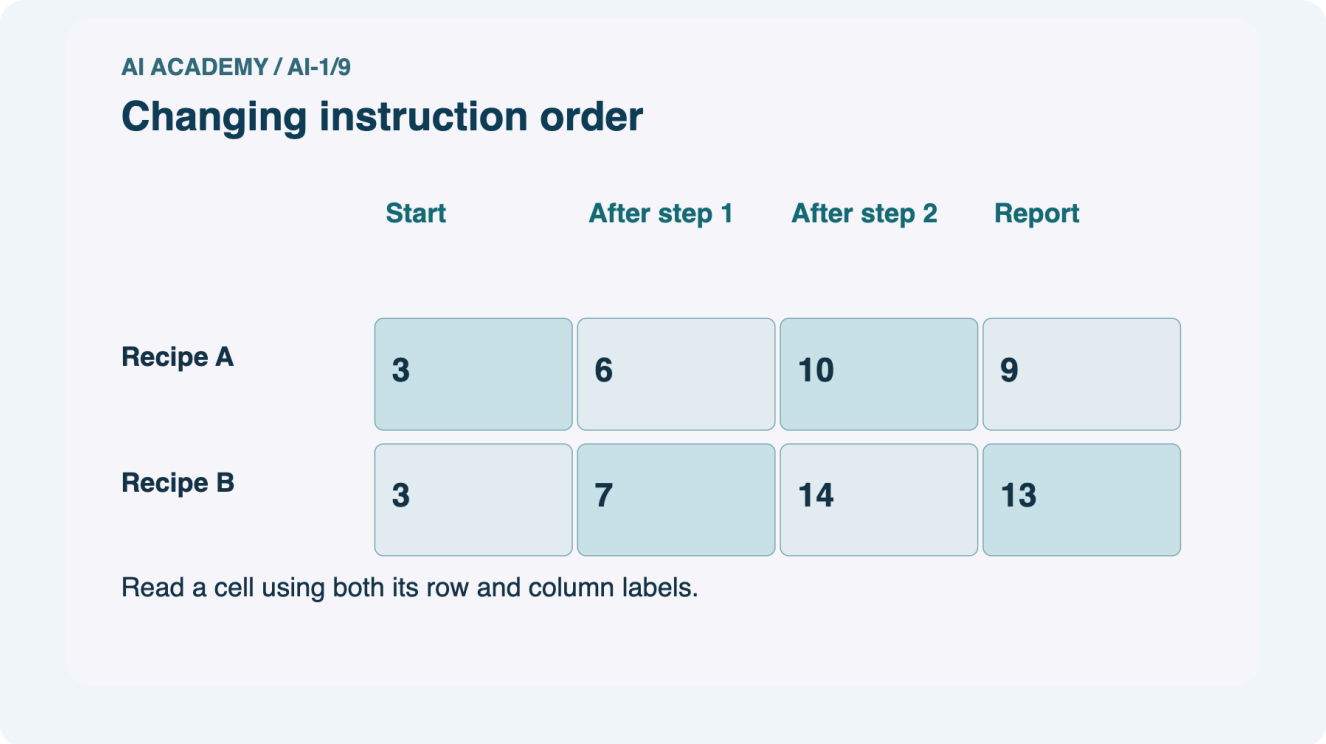
Start: Both runs begin at 3. → Recipe A: 3 → 6 after doubling → 10 after adding 4 → 9 after subtracting 1. → Recipe B: 3 → 7 after adding 4 → 14 after doubling → 13 after subtracting 1. → Compare **output**s: Same operations, different order: **output**s 9 and 13.

> Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Changing instruction order




Follow the labelled picture with the matching explanation. Open the vector diagram

Changing instruction order. Recipe A: Start = 3, After **step** 1 = 6, After **step** 2 = 10, Report = 9. Recipe B: Start = 3, After **step** 1 = 7, After **step** 2 = 14, Report = 13.

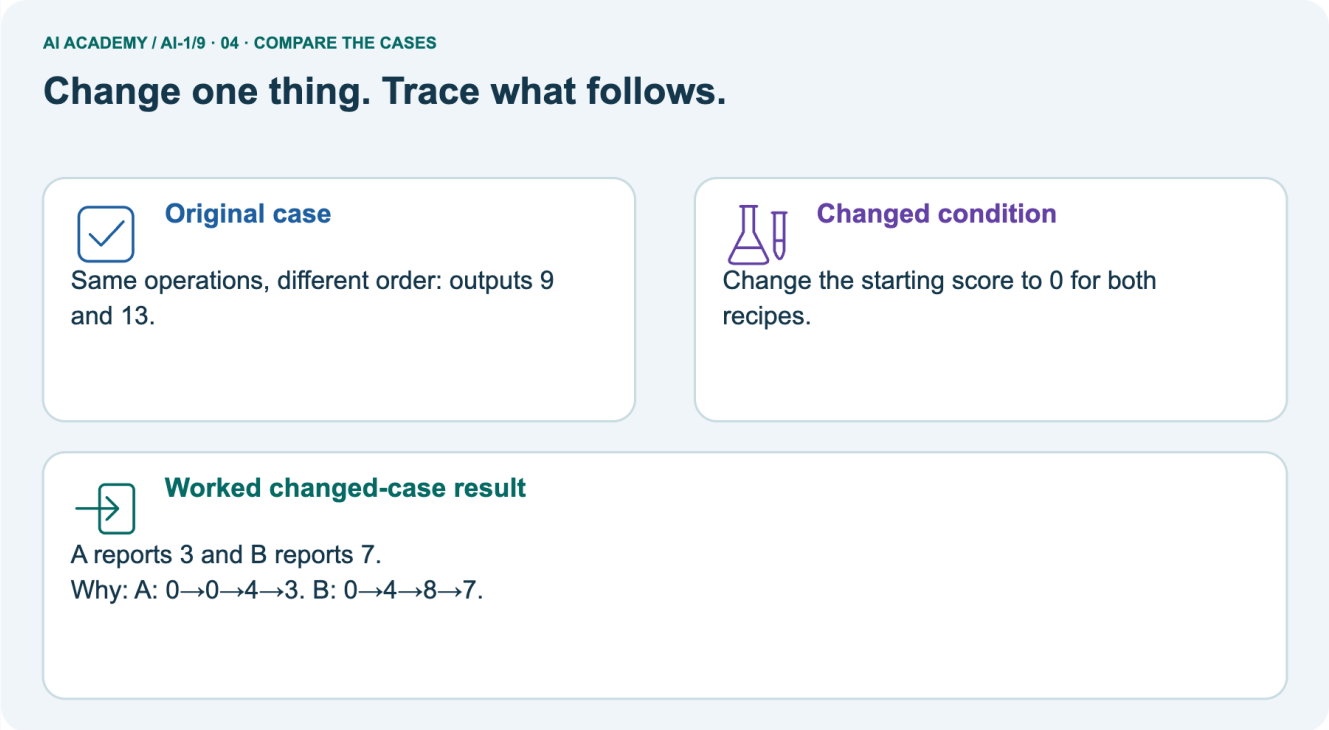
Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

**04. Change one thing. Trace what follows.**




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Same operations, different order: **output**s 9 and 13. Changed condition: Change the starting score to 0 for both recipes. Expected result: A reports 3 and B reports 7. Reason: A: 0→0→4→3. B: 0→4→8→7.

Predict before checking the worked changed-case result. Name what changed and what the

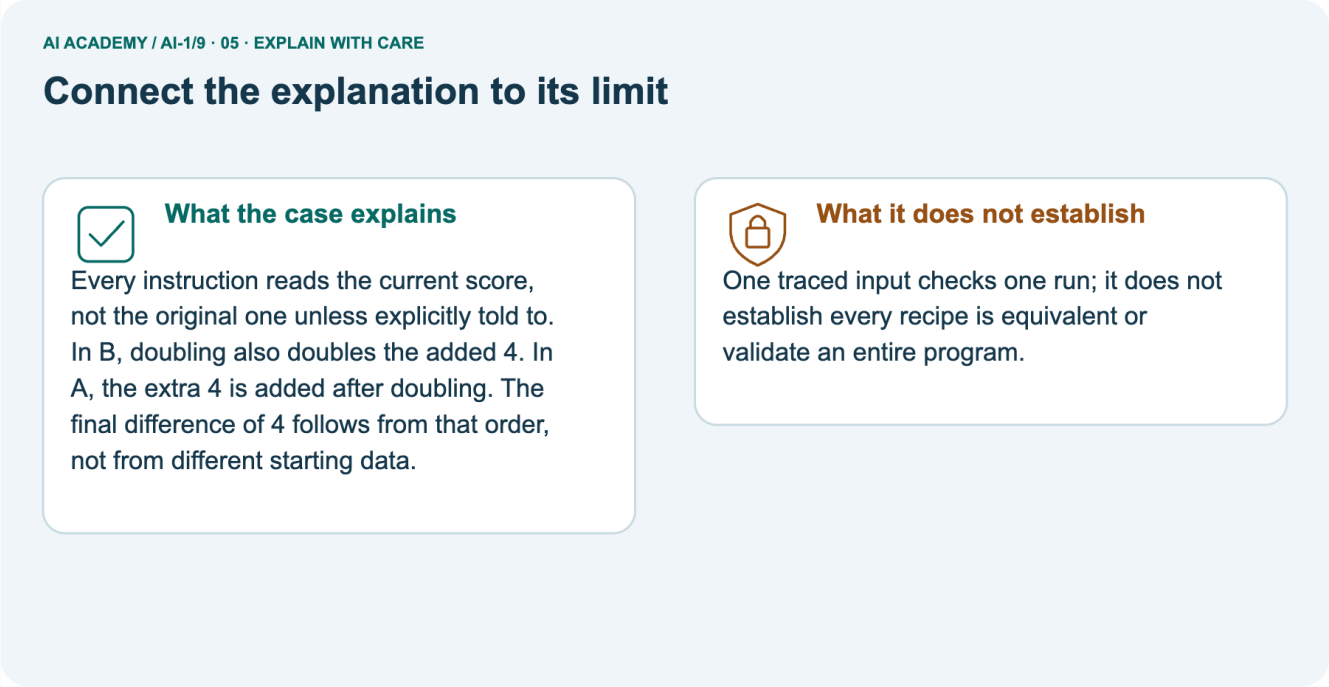
supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A: 0→0→4→3. B: 0→4→8→7.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Every instruction reads the current score, not the original one unless explicitly told to. In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling.

The final difference of 4 follows from that order, not from different starting data. Limit: One **trace**d **input** checks one run; it does not establish every recipe is equivalent or validate an entire program.

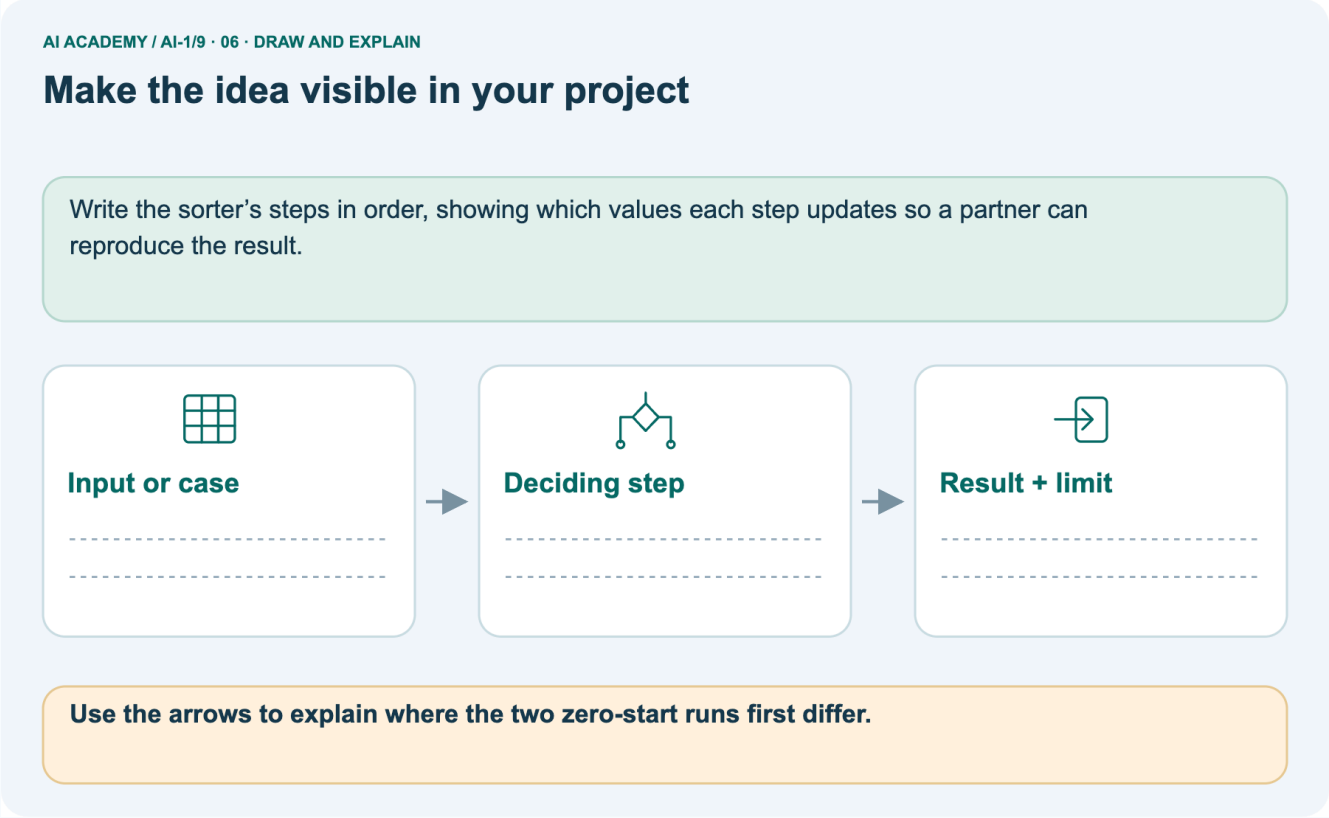
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the **state**d scope: One **trace**d **input** checks one run; it does not establish every recipe is equivalent or validate an entire program.

### 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result. Draw your **input**, deciding **step**, result and limit.

Supported question: Use the arrows to explain where the two zero-start runs first differ.

Draw your project **trace** and justify the deciding **step** using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project **trace** and justify the deciding **step** using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: They first differ after operation one: A has 0 and B has 4. Their reports are 3 and 7.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.